

DSCI 2000 Foundations of Data Science

Final Project (Fall 2026)

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Pathways to Scholarships

A Data-Driven Study of Scholarships for Middle and High School Students Across America

Project Overview

In this project, you will conduct a comprehensive study of scholarships available to middle and high school students across the United States. The project is intentionally restricted to scholarships: do not collect general enrichment programs, camps, clubs, competitions, internships, or other student opportunities unless they directly provide a monetary scholarship, educational award, or scholarship funding.

Each student will focus on the U.S. states assigned by the instructor and will also identify selected national scholarships. The class datasets will be combined into a nationwide scholarship dataset for exploratory analysis, visualization, comparison, and recommendations.

Project Goal

By the end of the project, the class will have created a structured dataset that helps students and parents understand what scholarships exist, who is eligible, how much funding is available, when applications are due, and what factors may affect access. The project will strengthen skills in data collection, data cleaning, exploratory data analysis, visualization, interpretation, and evidence-based communication.

Scope: What Counts as a Scholarship?

- Include scholarships or educational awards that provide money, tuition support, savings awards, or another clearly stated financial benefit to an eligible student.
- Include scholarships open to middle school students, high school students, or both.
- Include state/local scholarships and national scholarships available to U.S. students.
- Exclude ordinary camps, classes, clubs, competitions, internships, and enrichment programs unless a scholarship or monetary educational award is explicitly attached.
- Record the scholarship provider's official page. Scholarship-search websites may be used for discovery, but important details should be verified against the provider's current page before final submission.

Questions for Analysis

Here are 25 scholarship-focused questions for the class dataset. You must answer any 16 of the first 24 questions and also propose and answer Question 25.

1. Which states have the highest number of scholarships available to middle and high school students in our dataset?

2. What percentage of scholarships are open to middle school students, high school students, or both?
3. Which high school grade levels (9, 10, 11, or 12) have the greatest number of scholarship opportunities?
4. What is the distribution of scholarship award amounts, and what are the median and average award amounts?
5. What percentage of scholarships offer awards below \$1,000, \$1,000-\$4,999, \$5,000-\$9,999, and \$10,000 or more?
6. Which scholarships offer the largest awards, and what eligibility requirements are associated with those awards?
7. Are scholarship deadlines concentrated in particular months or seasons of the year?
8. How far in advance of the deadline do scholarships typically open for applications, when opening dates are available?
9. What are the most common eligibility requirements: GPA, grade level, residency, financial need, community service, leadership, academic achievement, talent, or other criteria?
10. What percentage of scholarships require an essay, and do essay-required scholarships tend to have larger award amounts?
11. What percentage require recommendation letters, transcripts, test scores, interviews, portfolios, videos, or other supporting materials?
12. How common are merit-based, need-based, community-service, leadership, STEM, arts, athletics, and other scholarship categories?
13. Which academic or career fields are most frequently supported by scholarships?
14. Are STEM scholarships more common or larger in value than scholarships in arts, humanities, business, health, or other fields?
15. What percentage of scholarships are restricted by state or local residency compared with scholarships open nationally?
16. Which states appear to have the greatest total scholarship dollars represented in the dataset?
17. Do scholarships with more restrictive eligibility requirements tend to offer larger awards?
18. How many scholarships are renewable for multiple years, and how does their value compare with one-time awards?
19. What percentage of scholarships are offered by foundations/nonprofits, corporations, colleges/universities, government agencies, professional associations, or community organizations?
20. How many scholarships are specifically targeted to students with financial need, and how do their award amounts compare with scholarships that do not consider financial need?
21. How many scholarships are accessible to students before senior year, and what does this suggest about when students should begin searching?
22. How many scholarships are open to middle school students, and how do their award types and amounts differ from high school scholarships?
23. Are there meaningful regional differences in scholarship availability, award amount, eligibility, or category across the Northeast, Midwest, South, and West?
24. Based on the data, what practical recommendations would you give students and parents for when, where, and how to search for scholarships?
25. What additional scholarship-related question do you want to ask and answer using the dataset?

Recommended Data Fields

The Excel template should capture enough fields to answer the research questions. Recommended fields include:

- Scholarship name
- Scholarship provider/organization
- Provider type
- Official scholarship URL
- Discovery source
- State/region
- National vs. state/local
- Eligible grade levels
- Middle school / high school / both
- Scholarship category
- Field/area of interest
- Award amount
- Number of awards (if known)
- Renewable (Yes/No/Unknown)
- Maximum potential value
- Application open date (if known)
- Application deadline
- Residency requirement
- Minimum GPA (if any)
- Financial need required (Yes/No/Unknown)
- Essay required
- Recommendation required
- Transcript/test score required
- Community service requirement
- Leadership requirement
- Other major eligibility requirements
- Application method
- Verification date
- Notes

Implementation Phases

The project is divided into four phases, each worth 50 points (total: 200 points).

Phase 1: Scholarship Data Collection (50 Points)

- Research scholarships available to middle and high school students in the U.S., focusing on the states assigned by the instructor.
- Collect approximately 10 verified scholarships per assigned state, where feasible, and 2-3 national scholarships.
- Use the provided Excel template; each row should represent one scholarship.
- Use scholarship databases for discovery, but verify award amount, deadline, and eligibility on the scholarship provider's official page whenever possible.
- Record a verification date so the class can distinguish current information from stale listings.
- If a state has very few scholarships for younger students, document that finding rather than substituting unrelated student opportunities.

Phase 2: Exploratory Data Analysis (EDA) (50 Points)

- Clean and standardize fields such as award amount, grade level, deadline, scholarship type, state, and eligibility.
- Calculate summary statistics and frequency distributions.
- Identify missing values, duplicate scholarships, expired listings, and inconsistent categories.
- Perform initial comparisons across states, grade levels, scholarship categories, award amounts, and deadlines.

Phase 3: Data Analysis and Visualization (50 Points)

- Create clear visualizations such as bar charts, histograms, box plots, maps, and deadline timelines.
- Use visualizations to compare scholarship availability, award amounts, categories, grade levels, eligibility requirements, and geographic patterns.
- Conduct deeper analysis guided by at least 16 of the 25 scholarship research questions.
- Summarize findings and provide practical recommendations for students, parents, counselors, and educators.

Phase 4: Analysis Summary and Final Presentation (50 Points)

- Present the most important findings, visualizations, state-level observations, and recommendations.
- Explain limitations of the dataset, including incomplete listings, changing deadlines, duplicated listings across websites, and scholarships that may require account-based searches.
- Presentations should be clear, professional, evidence-based, and useful to students and parents.

Recommended Scholarship Data Sources

Use multiple sources because no single scholarship database is complete. These sites are useful starting points for discovering scholarships. Middle-school-specific scholarships are less common, so students should also search local foundations, community organizations, school districts, state agencies, and the official sites of scholarship providers.

Source	Best Use	Website
BigFuture Scholarship Search (College Board)	Large scholarship directory with filters and high-school-focused college planning resources.	Visit site
Fastweb	Long-running scholarship search and matching database for high school and college-bound students.	Visit site
Scholarships.com	Scholarship directory with grade-level, state, deadline, amount, and other categories.	Visit site
Apply	Scholarship database with grade-level and eligibility filters, including high school freshmen through seniors.	Visit site
Going Merry	Scholarship search and application platform with resources for high school students and counselors.	Visit site

Niche Scholarships	Searchable scholarship listings including grade-specific opportunities for high school students.	Visit site
Bold.org	Scholarship platform containing many private awards and student eligibility categories.	Visit site
Unigo	Scholarship directory and college-planning resource with student-focused award listings.	Visit site
Scholarship America	Nonprofit scholarship organization that administers and lists scholarship programs.	Visit site
JLV College Counseling Scholarship Lists	Curated scholarship lists organized by deadline and student characteristics; useful as a discovery source.	Visit site

Data-quality rule: A listing on an aggregator is not enough by itself. Before final submission, verify the scholarship on the provider's official website whenever an official page is available. Record the official URL and the date verified.

Submission Requirements

- 1. Excel Data File:** Submit the completed scholarship dataset for your assigned states and national scholarships. Ensure required fields are filled accurately and sources are recorded.
- 2. EDA Notebook:** Submit a Jupyter/Colab notebook containing data cleaning, summary statistics, exploratory analysis, and responses to Phase 2 tasks.
- 3. Visualizations:** Include clear and well-labeled scholarship visualizations in the notebook or report.
- 4. Data Analysis & Summary Report:** Submit a 2-3 page written summary addressing at least 16 of Questions 1-24 plus your student-defined Question 25. Include recommendations and dataset limitations.
- 5. Final Presentation:** Prepare a PowerPoint, Google Slides, or similar presentation summarizing findings, visualizations, and recommendations. Record a 7-10 minute presentation video.
- 6. File Naming:** Include LastName_FirstName in all submission filenames (for example, Doe_John_Data.xlsx).
- 7. Deadlines:** Submit all deliverables by the dates specified on the course schedule. Late submissions are subject to deductions.